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PERFORMANCE SPECIFICATION

SWITCH, PROXIMITY, SOLID-STATE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for shipboard solid-state proximity switches. The switches use an inductive sensor for noncontact position sensing of metallic objects. The switches transfer power between a power source (see 6.4.13) and its loads when the switch is actuated.

1.2 Classification. Solid-state proximity switches are of the following designations, as specified (see 6.2):

1.2.1 Switch designations. Switches are designated according to type of input voltage (alternating current (AC) or direct current (DC)); maximum sensing distance (see 6.4.10) in inches; and output states (NO for normally open, NC for normally closed, and NONC for both normally opened and normally closed), as follows (see 6.6):

a. AC - 0.5 - NO
b. AC - 1.0 - NO
c. AC - 1.5 - NO
d. AC - 0.5 - NC
e. AC - 1.0 - NC
f. AC - 1.5 - NC
g. DC - 0.5 - NONC
h. DC - 1.0 - NONC
i. DC - 1.5 - NONC

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05M2, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-F-24385	-	Fire Extinguishing Agent, Aqueous Film-Forming Foam (AFFF) Liquid Concentrate, for Fresh and Sea Water

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-202	-	Electronic and Electrical Component Parts
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-704	-	Aircraft Electric Power Characteristics
MIL-STD-1399-300	-	Interface Standard for Shipboard Systems Section 300 Electric Power, Alternating Current (Metric)

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Switch characteristics.

3.2.1 AC switches, electrical characteristics. AC switches shall be two-wire devices and shall operate in series with the load. Each switch shall have one NO or NC solid-state output.

3.2.1.1 Operating voltage. The load and switches shall operate across a single-phase power source of 65 to 130 volts alternating current (VAC), 60-hertz (Hz). Operation for 115 VAC shall be in accordance with MIL-STD-1399-300, Type I (see 4.5.1).

3.2.1.2 Voltage drop. The voltage drop across each switch, during the activated (ON) state, shall be not greater than 4.0 VAC root mean square (rms) at rated load current (see 4.5.1).

3.2.1.3 Leakage current. The leakage current through each switch, during the deactivated (OFF) state, shall be not greater than 3.0 milliamperes (mA) rms (see 4.5.1).

3.2.1.4 Load ratings. The load rating of the switches shall operate as specified (see 4.5.1). The load ratings of the switches shall be as specified in 3.2.1.4.1 through 3.2.1.4.4.

3.2.1.4.1 Resistive. Switches shall have a resistive rating of 1.25 amperes rms.

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3.2.1.4.2 Inductive. Switches shall operate inductive loads with a power factor between 1 and 0.35. As a minimum, switches shall have a make-current rating of 10.0 amperes rms for three cycles of the specified operating voltage and a break current rating (see 6.4.2) of 1.25 amperes rms.

3.2.1.4.3 Minimum load. Switches shall operate with a minimum load of 15 mA rms.

3.2.1.4.4 Overload. Absolute overload and short circuit protection are desired. As a minimum, switches shall be protected from continuous overloads which would result in up to 6.0 amperes rms at normal output voltage. For devices which utilize a change of state of the switch from normal to overload, interruption of the operating voltage to restore normal operation of the switch after an overload has been detected is permissible. For devices which provide overload protection by current limiting as the protection logic, normal system performance shall be automatically resumed when the overload condition is rectified.

3.2.1.5 Multiple switch operation. Multiple switch operation shall be as specified (see 4.5.1).

3.2.1.5.1 Series connection. Switch designations shall operate with two identical switches in series with a power source of 65 to 130 VAC. Switch designations shall operate with three identical switches in series with an operating voltage of 115 VAC in accordance with MIL-STD-1399-300. The switch voltage drop shall not be greater than 4.0 VAC rms per switch at the rated load current.

3.2.1.5.2 Parallel connection. Switch designations shall operate with two identical switches in parallel. The total switch leakage current through the load shall be not greater than 4.0 mA rms.

3.2.1.6 Transients. Switches shall continue to operate without degradation when the operating power or output is subjected to voltage and frequency transients in accordance with MIL-STD-1399-300 for 115 VAC, Type I power (see 4.5.1). The spike voltage test amplitude shall be 1000 volts. The spike waveform shall be in accordance with MIL-STD-1399-300. Application of a deviation in voltage over a deviation in time (dv/dt) of 1000 volts per microsecond across the output contact leads at the end of the supplied cable shall not result in false closure or opening of the output.

3.2.1.7 AC wire colors. Wire colors shall be as follows:

NO switches:	NC switches:
Black – Input	Black – Input
White – Output	White – Output

3.2.2 DC switches, electrical characteristics. DC switches shall be four-wire devices and shall operate as voltage sources. Switches shall be line powered and shall have solid-state outputs. Each switch shall have one NO and one NC output in a complementary configuration. The total power drawn from the line shall be not greater than the sum of 50 mA plus the output load current.

3.2.2.1 Operating voltage. DC switches shall operate with any voltage between 10 and 30 volts of continuous current (Vdc) (see 4.5.1).

3.2.2.2 Voltage drop. The voltage drop across each switch, during the activated (ON) state, shall be not greater than 1.5 Vdc at rated load current (see 4.5.1).

3.2.2.3 Leakage current. Leakage current from each output in the open (OFF) state shall be not greater than 300 microamperes with 35 Vdc applied to the switch as specified (see 4.5.1).

3.2.2.4 Load ratings. The switch shall operate as specified with the load ratings of the switch as specified in 3.2.2.4.1 through 3.2.2.4.3 (see 4.5.1).

3.2.2.4.1 Maximum current. The switches shall continuously supply 250 mA to resistive and inductive loads and 100 mA into lamp loads. The inductive load shall have a maximum decay time of 100 milliseconds when the load is interrupted. This decay shall be measured from the 90 percent level to the 10 percent level.

3.2.2.4.2 Minimum load. The switches shall supply any load current from maximum down to zero.

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3.2.2.4.3 Overload. The switches shall be protected from damage due to an overload or direct short circuit of the output. For devices which utilize a change of state of the switch from normal to overload, interruption of the operating voltage to restore normal operation of the switch after an overload has been detected is permissible. For devices which provide overload protection by current limiting as the protection logic, normal system performance shall be automatically resumed when the overload condition is rectified.

3.2.2.5 Reverse polarity protection. Switches shall be immune to damage from reversal of the input power or output leads (see 4.5.1).

3.2.2.6 Transients. Switches shall continue to operate without degradation when the input power and outputs are subjected to voltage transients in accordance with MIL-STD-704 for 28-volt systems (see 4.5.1).

3.2.2.7 Dielectric withstand. Switches shall withstand a ±250 Vdc potential between switch case and each individual input and output connection for 60 seconds with less than 0.75 mA rms leakage current flow. Measurement shall be in accordance with MIL-STD-202, Method 301.

3.2.2.8 Insulation resistance. Insulation resistance between the switch case and each individual input or output lead or operational power lead shall be greater than 10 megohms. Measurement shall be in accordance with MIL-STD-202, Method 302, test condition A for a duration of 2 minutes. The test may be terminated before the end of 2 minutes if the measured insulation resistance is greater than 10 megohms and is steady or increasing.

3.2.2.9 DC cable wiring. Wire colors shall be as follows:

a. Black – DC power high (plus lead)

b. White – NO output

c. Red – NC output

d. Green – DC power low (minus lead)

3.2.2.10 Multiple switch operation. Multiple switch operation shall be as specified (see 4.5.1).

3.2.2.10.1 Series connection. Switch designations shall operate with two identical switches in series. The total switch voltage drop at the load shall be not greater than 3.0 Vdc at rated load current.

3.2.2.10.2 Parallel connection. Switch designations shall operate with two identical switches in parallel. The total switch leakage current through the load shall be not greater than 600 microamperes.

3.2.3 Operation (see 6.4.7).

3.2.3.1 Measurements. Measurements shall be made using standard targets (see 6.4.14) as specified in 3.2.3.4.

3.2.3.2 Head-on mode (see 6.4.5).

3.2.3.2.1 Sensing distance (see 6.4.10). The sensing distances for the switches shall be as follows (see 4.5.1):

a. For all 1.5-inch switches, the sensing distance in the head-on mode shall be 1.50 plus 0.50 minus 0.12 inches.

b. For all 1.0-inch switches, the sensing distance in the heads on mode shall be 1.0 plus 0.22 minus 0.08 inch.

c. For all 0.5-inch switches, the sensing distance in the head-on mode shall be 0.5 plus 0.05 minus 0.05 inch.

The sensing distance of the switches shall operate as specified (see 4.5.1).

3.2.3.2.2 Hysteresis. Hysteresis (see 6.4.6) distances for the switches in the head-on mode shall be as follows (see 4.5.1):

a. For all 1.5-inch switches, a minimum of 5 percent and a maximum of 40 percent of the nominal sensing distance.

b. For all 1.0-inch switches, a minimum of 5 percent and a maximum of 30 percent of the nominal sensing distance.

c. For all 0.5-inch switches, a minimum of 5 percent and a maximum of 20 percent of the nominal distance.

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3.2.3.3 Slide-by mode (see 6.4.12).

3.2.3.3.1 Repeatability to actuation curve. The sensing distance in the slide-by mode (see figure 1) at any distance from the sensing face (see 6.4.11) shall vary not more than ±0.12 from the value on the contractor's actuation curve. The repeatability (see 6.4.9) shall be true over the specified voltage and temperature ranges. The repeatability shall also be true at any distance between 0.0 and 1.12 inches for all 1.5-inch switches (see figure 2), between 0.0 and 0.75 inch for all 1.0-inch switches (see figure 3), and between 0.0 to 0.38 inch for all 0.5-inch switches from the sensing face (see figure 4).

3.2.3.3.2 Actuation curve slope requirements. The slope of the actuation curve shall be as steep as possible (see figures 2, 3, and 4). Slope of the actuation curve for 1.5-inch switches shall not be less than 45 degrees between 0.5 and 0.75 inch above the sensing face in the slide-by mode while moving the standard target in planes parallel to the sensing face. This slope shall be determined from measurements made at 0.25 and 0.75 inch above the sensing face.

3.2.3.3.3 Hysteresis. Hysteresis distances for the switches in the slide-by mode shall be as follows (see 4.5.1):

a. For all 1.5-inch switches, a minimum of 0.025 inch and maximum of 1.12 inches in parallel planes up to 1.12 inches above the sensing face.

b. For all 1.0-inch switches, a minimum of 0.017 inch and a maximum of 1.0 inch in parallel planes up to 0.75 inch above the sensing face.

c. For all 0.5-inch switches, a minimum of 0.01 inch and a maximum of 0.5 inch in parallel planes up to 0.38 inch above the sensing face.

3.2.3.4 Standard target. The target shall be 4130 steel, 0.062 inch thick in the fully annealed condition and shall require no special treatment. The target shall not be active in any manner or magnetization. The complimentary proximity sensor (see 6.4.3) shall respond to the target bulk material properties and not to any surface finish. For all 1.5-inch switches, the target shall be 4 inches square. For all 1.0- and 0.5-inch switches, the targets shall be 3 inches square. Target thickness shall be not less than 0.062 inch and the thickness shall be uniform. These targets shall be used to demonstrate conformance to this specification. Other target shapes, sizes, and other common grade annealed ferrous metals may be used in service.

3.2.3.5 Response time. The switches shall change operating state within 50 milliseconds after the standard target is removed or inserted for any load conditions (see 4.5.1).

3.2.3.6 Adjacent metals. The switches shall operate as specified herein in the presence of metal against the back of the case or against the side of the case up to the plane of the sensing face. Metal at the level of the sensing face or above shall be at least one-half the sensing face diameter away from the edge of the sensing face (see 4.5.1).

3.2.3.7 Facing metals. The switches shall operate as specified herein in the presence of ferrous metal directly in front of the sensor face as close as three times the sensing distance. For nonferrous metals, the distance may be as close as two times the sensing distance (see 6.4.10).

3.2.3.8 Adjacent sensors. Multiple sensors shall operate as specified herein when placed side by side with the distance between adjacent edges of the switches greater than the sensing distance. Sensors shall operate as specified herein when placed face to face with the distance between sensing faces greater than six times the sensing distance (see 4.5.1).

3.2.3.9 Magnetic attraction. The sensor shall not establish a magnetic field so strong as to attract or hold dust size or larger ferromagnetic materials to the sensor face. Dust size particles shall be standard reagent quality iron powder (100 to 200 mesh) (see 4.5.1).

3.2.3.10 Switch stabilization. Switches shall stabilize to normal operation within 500 milliseconds of initial application of operating power. False output transitions shall not occur during power up when the minimum dv/dt of the applied voltage is greater than 200 volts per second (see 4.5.1).

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3.2.4 Physical characteristics.

3.2.4.1 Weight. The weight of each switch, excluding the cable, shall be not greater than 7.0 pounds (see 4.3.3).

3.2.4.2 Dimensions. The dimensions shall conform to figure 5 for 1.0-inch and 1.5-inch switches and figure 6 for 0.5-inch switches.

3.2.4.3 Cable. The contractor shall supply a resilient, UL-listed cable with each switch (see 6.5.1). The cable shall be 20 feet in length. If shielded cable is selected, the shield shall not be electrically connected to the switch case. The cable shall be selected based on previously specified color coding, adequate electrical characteristics, and the following physical properties: weatherability, moisture resistance, cold flexibility, flame retardancy, resistance to mechanical abuse, aqueous film-forming foam (AFFF) compatibility, and chemical inertness. Conductors shall be stranded. The cable outside diameter shall be not greater than 0.5 inch. The cable shall be strain-relieved and sealed into the switch (see 4.3.3).

3.2.4.4 Encapsulation. The internal electronic circuitry and cable entry shall be encapsulated (see 4.3.3). The contractor shall select an appropriate compound which will enhance the ability of the switch to meet all requirements of this specification (see 6.5.2).

3.2.5 Reliability. Switches shall have a mean-time-between-failure (MTBF) of not less than 45,000 hours or 500,000 actuation cycles at full rated load (see 4.5.2).

3.2.6 Maintainability. Switches shall be nonrepairable and nonadjustable as a final item.

3.2.7 Environmental factors.

3.2.7.1 Temperature. Switches shall operate within specified accuracy in ambient temperature ranging from -4 to +158 °F (-20 to +70 °C) (see 4.5.3).

3.2.7.2 Humidity. Switches shall operate within specified accuracy in accordance with MIL-STD-202, Method 103B, test condition B (see 4.5.3).

3.2.7.3 Vibration. Switches shall withstand Type I vibration in accordance with MIL-STD-167-1. Vibration test frequency shall be from 4 to 50 Hz, inclusive. There shall be no electrical or mechanical damage to the switch and no momentary change of output states as a result of vibration testing (see 4.5.4).

3.2.7.4 Shock. Switches shall meet the shock requirements in accordance with MIL-S-901 for Grade A, Type A, Class I, lightweight equipment (see 4.5.5). Switches shall be operational during and after the shock test. No portion of the switch shall break off or otherwise create a hazard to personnel or other components. There shall be no momentary change of output states. Undesired changes of output state shall be plotted and represented graphically in the test record.

3.2.7.5 Salt atmosphere (corrosion). Switches shall operate after being subjected to the salt atmosphere test in accordance with MIL-STD-202, Method 101E (see 4.5.6).

3.2.7.6 AFFF compatibility. Switches shall be compatible with the AFFF in accordance with MIL-F-24385 (see 4.5.8). Compatibility shall be with both liquid and foam AFFF.

3.3 Materials. Materials shall be as specified (see 4.3.3).

3.3.1 Case and mounting plate. The contractor shall select an appropriate corrosion-resistant material for the switch case and mounting plate, if used (see 6.5.3). If the material is a nonmetal, it shall be resistant to damage from asymmetric mechanical loading. Painting or other surface treatment shall not be required to provide corrosion-resistance. Surface shall not require maintenance.

3.3.2 Prohibited materials.

3.3.2.1 Asbestos. Asbestos shall not be used.

3.3.2.2 Cadmium. Cadmium shall not be used.

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3.3.2.3 Mercury. Mercury in any form shall not be used in shipboard equipment, including materials and parts thereof. Mercury shall not be used in manufacturing and test processes (including test equipment such as mercury indicators) applying to the basic equipment, but may be used in the manufacturing and test processes for materials and parts provided it is used so the materials and parts themselves cannot be contaminated. No instruments containing mercury shall be used in the manufacture or testing of any equipment destined for installation on a nuclear powered ship.

3.4 Electromagnetic compatibility. Switches shall conform to the most current revision of MIL-STD-461 for surface ships and metallic hull. Compliance with the following methods as a minimum is required: CE102, RE101, RE102, CS101, CS114, CS116, RS101, and RS103.

The following changes to the limits specified in MIL-STD-461 are permitted:

a. Method CS101: The applicable limit may be relaxed by 5dBuV.

b. Method CS114: The applicable limit may be relaxed by 10dBuA.

c. Method CS116: A relaxation of the pulse current requirement to 0.25 Imax may be allowed at 30MHz.

d. Method RE101: The applicable limit may be relaxed by 10dBpt.

e. Method RS103: The switches shall not exhibit any malfunction, degradation of performance, or deviation from specified indications when subjected to square wave modulated radiated electric fields with a maximum strength of 200 volts per meter for above deck usage and 10 volts per meter for below deck usage.

3.5 Identification and marking.

3.5.1 Identification label. The contractor shall provide a label on the back (the side facing the mounting surface) of each switch. The label shall not be deteriorated by exposure to salt water, glycol, petroleum products, or salt water AFFF solution in accordance with MIL-F-24385. As a minimum, the label shall contain the following information:

a. Identification as a proximity switch

b. Classification designation

c. Manufacturer’s code or part number traceable to date of manufacture

d. Serial number

e. National stock number (NSN)

f. Wiring schematic for hook-up of end function of wires by color code

g. Note stating “DO NOT MEGGER”

h. Manufacturer’s name and address

3.5.2 Marking requirements. Identification as a proximity switch and the NSN shall be marked on the switch so as to be readily visible when the switch is mounted. Minimum size of characters shall be 0.125 inch in height. Marking shall be accomplished by raised or depressed letters not less than 0.02 inch in height or depth.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. The items designated for first article inspection (see 4.2.1) shall be subjected to the examinations and tests as specified in table I.

4.2.1 First article sample. One switch of the designation submitted (see 1.2.1) shall be subjected to first article inspection.

4.3 Conformance inspection. Unless otherwise specified (see 6.2), conformance inspection shall consist of group A examinations and tests as specified in table I (see 4.3.3).

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4.3.1 Inspection lot. An inspection lot shall consist of all switches of the same designation, produced under essentially the same conditions, and offered for inspection at one time.

4.3.2 Sampling. Sampling is not required (see 4.3.3).

TABLE I. First article and conformance inspections.

Inspection or test	Requirement	Test method	First article	Conformance inspection group A
Operating voltage	3.2.1.1 and 3.2.2.1	4.5.1	X	X
Voltage drop	3.2.1.2 and 3.2.2.2	4.5.1	X	X
Current leakage	3.2.1.3 and 3.2.2.3	4.5.1	X	X
Load ratings	3.2.1.4 and 3.2.2.4	4.5.1	X	X[1/]
Sensing distance	3.2.3.2.1	4.5.1	X	X
Multiple switch operation	3.2.1.5	4.5.1	X	
Transients	3.2.1.6 and 3.2.2.6	4.5.1	X	
Reverse polarity protection	3.2.2.5	4.5.1	X	
Hysteresis	3.2.3.2.2	4.5.1	X	X
Response time	3.2.3.5	4.5.1	X	
Repeatability	3.2.3.3.1	4.5.1	X	
Adjacent metals	3.2.3.6	4.5.1	X	
Facing metals	3.2.3.7	4.5.1	X	
Adjacent sensors	3.2.3.8	4.5.1	X	
Magnetic attraction	3.2.3.9	4.5.1	X	
Switch stabilization	3.2.3.10	4.5.1	X	
Weight	3.2.4.1	4.3.3	X	X
Dimensions	3.2.4.2	4.3.3	X	X
Cable	3.2.4.3	4.3.3	X	X
Encapsulation	3.2.4.4	4.3.3	X	X
Materials	3.3	4.3.3	X	X
Reliability	3.2.5	4.5.2	X	
Electromagnetic compatibility	3.4	4.5.7	X	
Temperature	3.2.7.1	4.5.3	X	
Humidity	3.2.7.2	4.5.3	X	
Vibration	3.2.7.3	4.5.4	X	
Shock	3.2.7.4	4.5.5	X	
Salt atmosphere	3.2.7.5	4.5.6	X	
AFFF compatibility	3.2.7.6	4.5.8	X	

NOTE:
[1/] Resistive load only.

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4.3.3 Production inspection (group A). Unless otherwise specified (see 6.2), production inspection shall be made on every item offered for delivery. This inspection shall include the examinations and tests which will assess the workmanship and reveal omissions and errors of the production process. Examples of tests are: functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls and adjustments, and tests which detect hidden defects of materials. Unless otherwise specified (see 6.2), production inspection shall include the examinations and tests shown in group A of table I and shall be performed as specified in 4.5.

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.5.

4.5 Test methods.

4.5.1 Operational test. Operational tests shall be performed to verify electrical and operational performance characteristics and shall meet the requirements specified herein. Measurements shall be made using standard targets as specified in 3.2.3.4.

4.5.2 Accelerated life test. Sample switch shall be subjected to 720 hours of continuously cycled operation in 158 °F (70 °C) ambient at full rated load on each output. The switching rate shall be ten actuation cycles per minute. Reliability shall be as specified in 3.2.5.

4.5.3 Temperature and humidity tests. Temperature and humidity tests shall be conducted in accordance with MIL-STD-202 and switches shall be tested as specified in 3.2.7.1 and 3.2.7.2.

4.5.4 Vibration. The switch shall be subjected to the Type I vibration test in accordance with MIL-STD-167-1 and test shall be as specified in 3.2.7.3. Switch circuits shall remain energized and appropriately loaded throughout the test. A monitoring system capable of sampling at a rate of 100 samples per second or more shall be used to monitor switch outputs throughout the test. Any undesired changes of output state shall be plotted and represented graphically in the test record. For each part of the vibration test, test time shall be divided equally, with the test assembly and switch configured with target and without target present (that is, switch actuated and deactuated). Failure to conform to 3.2.7.3 and 4.5.4 shall be cause for rejection.

4.5.5 Shock. The switch shall be subjected to the high-impact shock test for Grade A, Type A, Class I, lightweight equipment in accordance with MIL-S-901 and shall be as specified in 3.2.7.4. Switch circuits shall remain energized and appropriately loaded throughout the test. A monitoring system capable of sampling at a rate of 100 samples per second or more shall be used to monitor switch outputs throughout the test. Test shall be repeated with target and without target present (that is, switch actuated and deactuated), for a total of eighteen shock blows. Failure to conform to 3.2.7.4 shall be cause for rejection.

4.5.6 Salt atmosphere (corrosion). Switches shall be tested in accordance with Method 101, test condition A of MIL-STD-202 and shall be as specified in 3.2.7.5. Evidence of excessive corrosion that will cause a failure of the switch shall be cause for failure.

4.5.7 Electromagnetic compatibility. Switches shall be tested in accordance with MIL-STD-461. In addition, normal operation shall be demonstrated in pulsed and continuous wave radio frequency fields throughout the frequency ranges of 2 to 32 MHz, 108 to 400 MHz, and 2 to 18 gigahertz, with the attached cables exposed and the switch housing connected to the circuit ground. Switches shall meet the requirements specified in 3.4.

4.5.8 AFFF compatibility. Switches shall be immersed to one-half of their length in a bath of AFFF for 96 continuous hours. The bath shall be a solution of 6 percent AFFF in a 5 percent concentration of salt water. The solution and ambient temperature shall be 95 °F (35 °C). The solution shall be kept agitated enough to keep the unimmersed part of the switch covered with foam. The AFFF compatibility shall be as specified in 3.2.7.6. After immersion, switches shall be cycled a minimum of once per minute for 10 minutes. If the switch does not open and close for any of the cycles, it shall constitute a failure.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The proximity switches are for use in position sensing applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Designation of switches required (see 1.2).

c. If required, the specific issue of individual documents referenced (see 2.2.1).

d. When first article inspection is required (see 3.1).

e. If conformance inspection is other than specified (see 4.3).

f. Sampling basis for production inspection (group A) if other than specified (see 4.3.3).

g. Examinations and tests for production inspection (group A) if other than specified (see 4.3.3).

h. Packaging requirements (see 5.1).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. Terms used in this specification are defined in 6.4.1 through 6.4.14 and graphically shown in figure 1.

6.4.1 Actuation point. The actuation point is the point at which the target approaching the sensing face is detected.

6.4.2 Break current rating. Break current rating is the steady-state current flowing in a device at the time current is interrupted.

6.4.3 Complementary proximity sensor. A complementary proximity sensor features a normally open and a normally closed output, both of which can be used simultaneously.

6.4.4 Deactuation point. The deactuation point is the point at which the target leaving the sensing zone is no longer detected.

6.4.5 Head-on mode. Head-on mode indicates when the actuator is parallel and centered on the sensitive face of the sensor, and its path of travel is perpendicular to the sensitive face along the reference axis (see 6.4.8). The performance criteria are given as dimensions from the sensing face to the leading edge of the actuator.

6.4.6 Hysteresis. Hysteresis is the distance between the actuation point (see 6.4.1) and the deactuation point (see 6.4.4) measured in the direction of target travel. In the head-on mode, it is measured along the reference axis. In the slide-by mode, it is measured in a plane parallel to the sensing face along a line perpendicular to the reference axis passing through the reference axis in the direction of target travel.

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6.4.7 Mode of operation. Mode of operation is the direction of actuator travel in relation to the sensing face of the sensor.

6.4.8. Reference axis. The reference axis is perpendicular to the sensing face along the reference axis (see 6.4.9).

6.4.9. Repeatability. Repeatability is the maximum shift in the sensing distance when the switch is operated anywhere within specified parameter limits.

6.4.10 Sensing distance. Sensing distance is the perpendicular distance from the actuation point to the sensing face in the head-on mode. It is the perpendicular distance from the actuation point to the reference axis in the slide-by mode.

6.4.11 Sensing face. The sensing face is a surface of the proximity switch which reacts to an alternating electromagnetic field.

6.4.12 Slide-by mode. Slide-by mode indicates when the actuator and its path of travel are parallel to the sensing face of the sensor. The performance criteria are given as dimensions from the reference axis to the leading edge of the actuator.

6.4.13 Source. Source is the device which switches a positive voltage to a load. The current flows through the sensor and into the load.

6.4.14 Target. Target refers to the item, material, or configuration that a sensor is intended to detect.

6.5 Application guidance notes.

6.5.1 Cable recommendations. Omniguard 2, Dekoron TPE on Plenum cable, or Belden Plenum cables have demonstrated physical properties appropriate to this specification. Other cables, including NSN 6145-00-110-2262, may be equally appropriate.

6.5.2 Potting compound recommendations. Rigid epoxy in accordance with MIL-I-16923, self-extinguishing, is an example of a suitable compound for encapsulation.

6.5.3 Casing material recommendations. Stainless steel (316) has attractive chemical and physical properties for the switch casing mounting plate, if used.

6.6 Switch designations. Switch designations (see 1.2.1) are constructed as follows:

XX – Y.Y – ZZZZ

XX - Type of input voltage

Y.Y - Maximum sensing distance

ZZZZ - Output states

6.7 Subject term (key word) listing.

Aqueous film forming foam

Head-on mode

Hysteresis

Inductive sensor

Sensing face

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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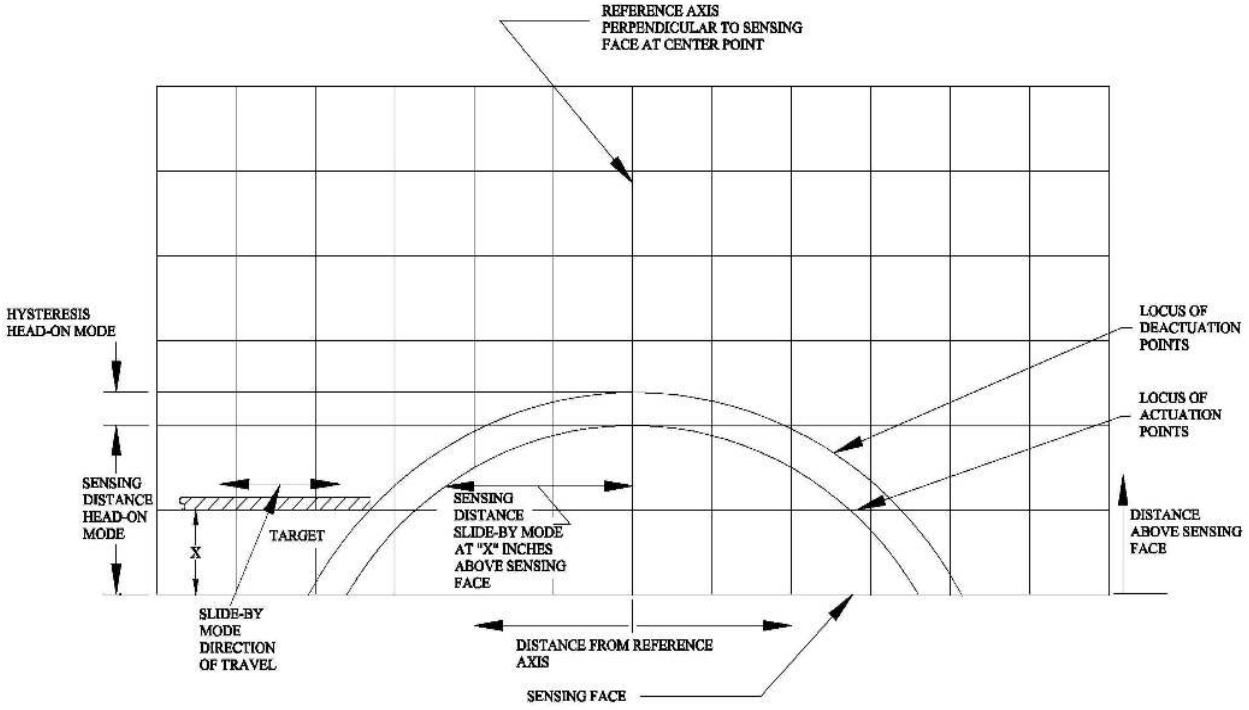


FIGURE 1. Typical actuation curve with hysteresis.

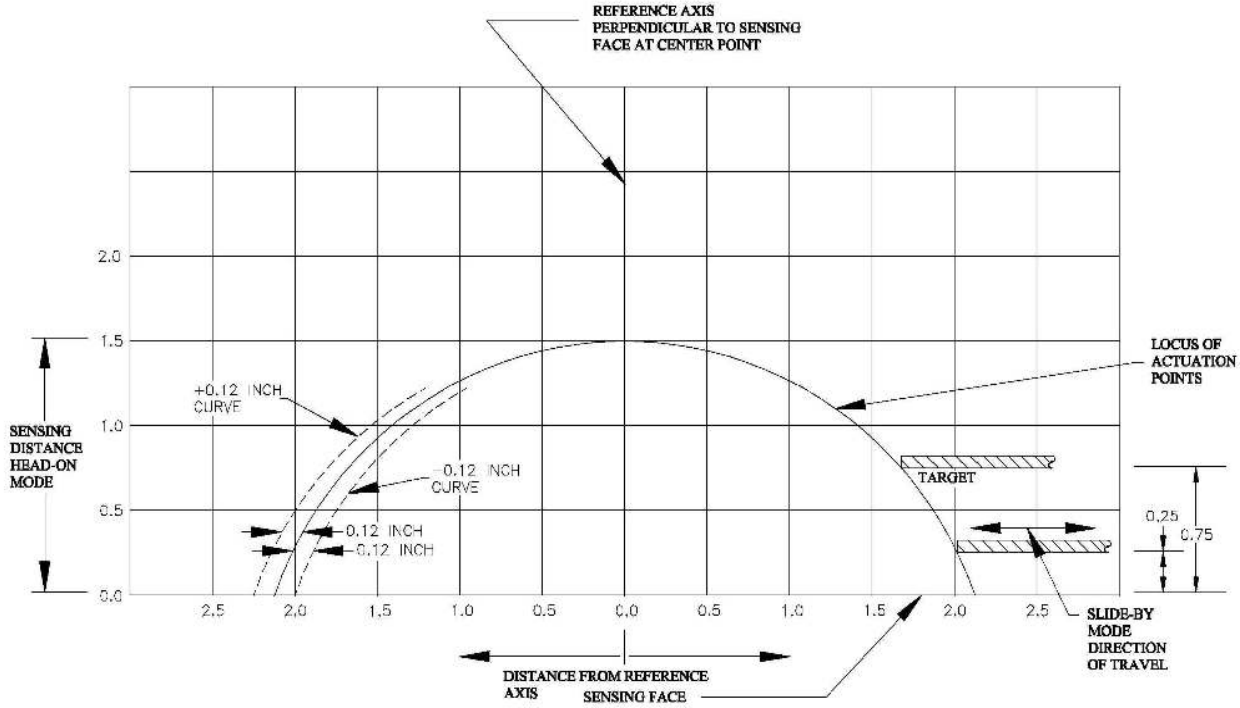


FIGURE 2. Repeatability curve slide-by mode, 1.5-inch switch.

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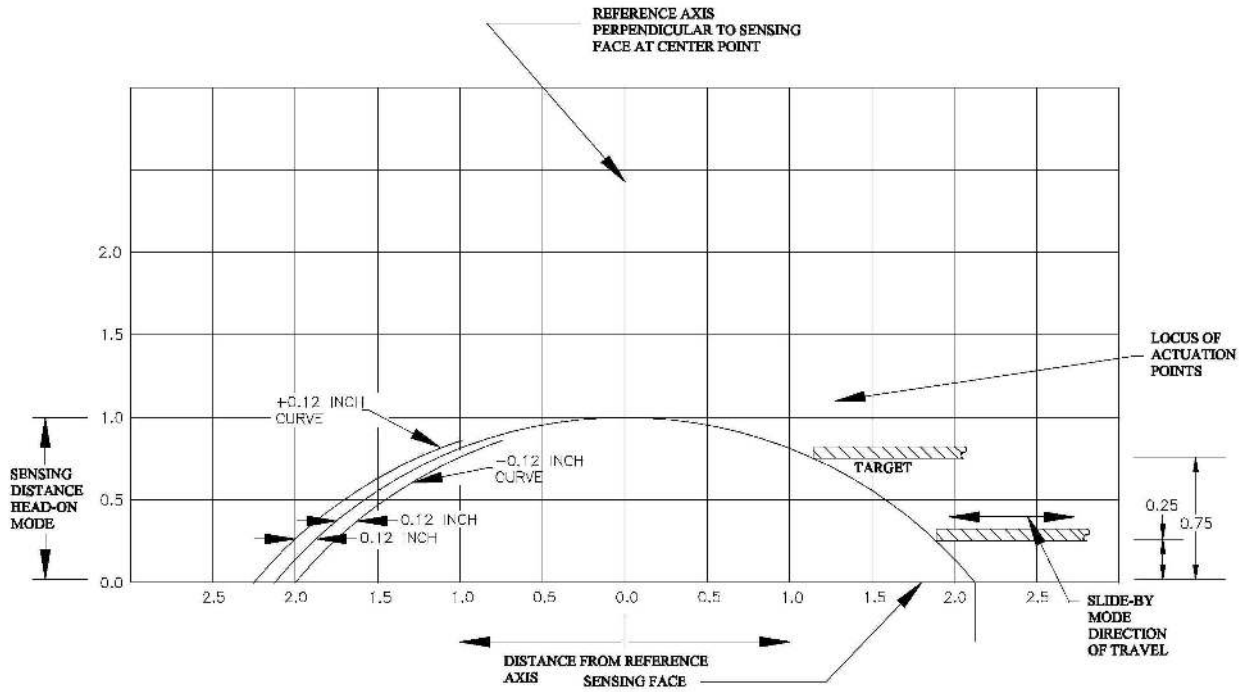


FIGURE 3. Repeatability curve slide-by mode, 1.0-inch switch.

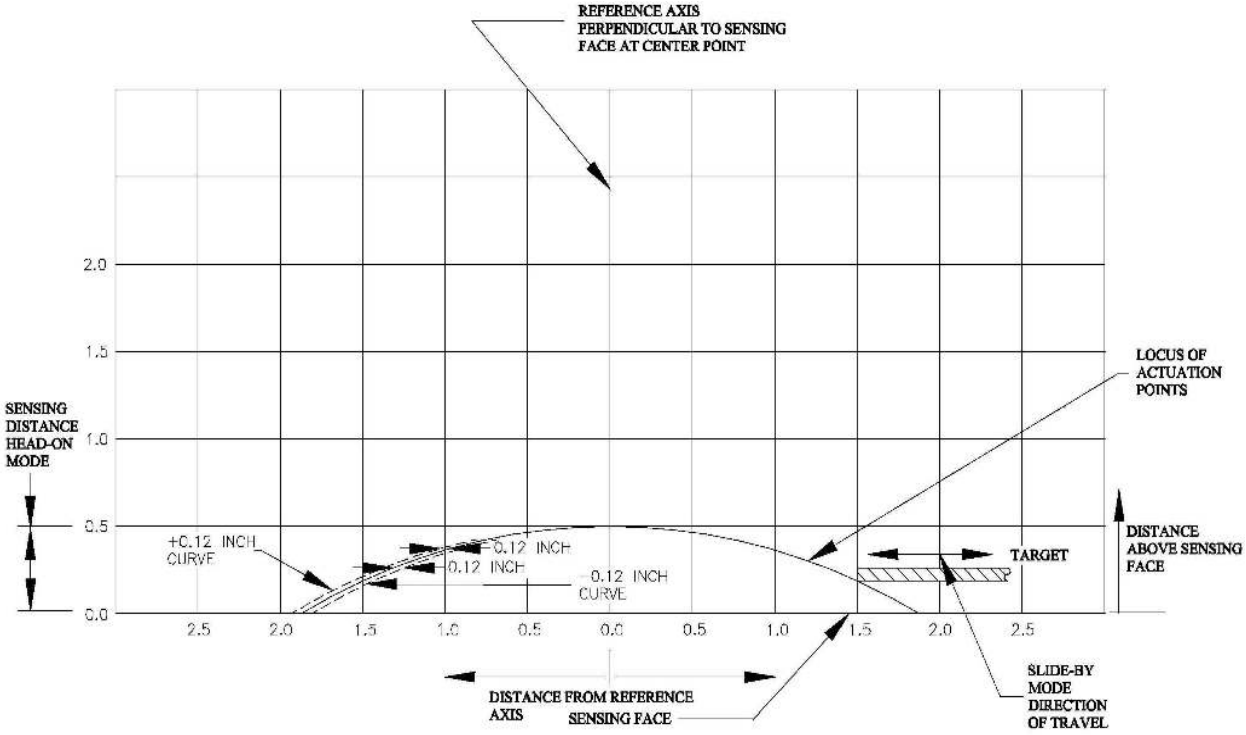


FIGURE 4. Repeatability curve slide-by mode, 0.5-inch switch.

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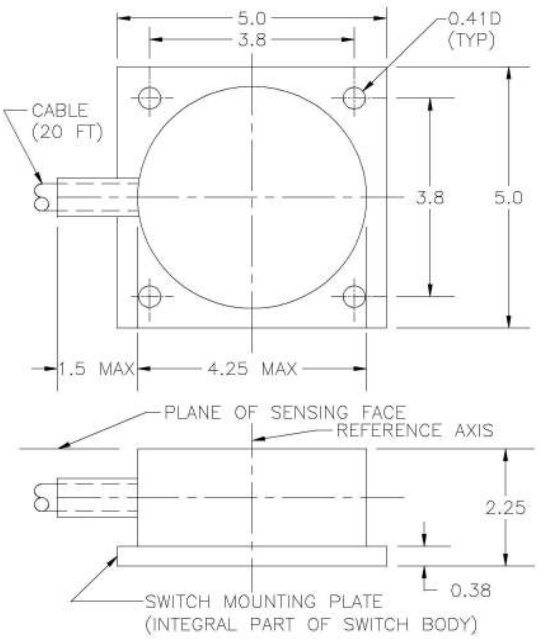


FIGURE 5. Outline dimensions for AC-1.0-NO, AC-1.5-NO, AC-1.0-NC, AC-1.5-NC, DC-1.0-NONC, DC-1.5-NONC switch.

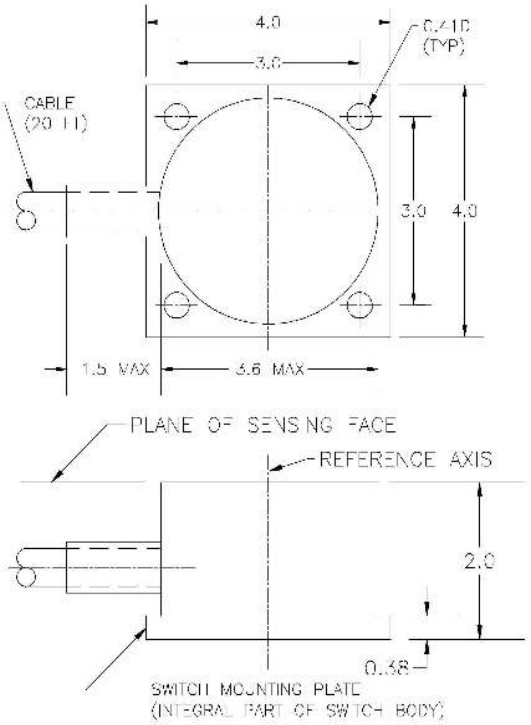


FIGURE 6. Outline dimensions for AC-0.5-NO, AC-0.5-NC, DC-0.5 NONC switch.

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